Operating Systems Course

Project Cover Sheets

Algorithm Comparison Projects

Student-facing cover sheets and submission guide for the five approved Part 1 comparison projects.

Included in this document:
- one cover sheet template per project
- one separate variant-specific checklist page for each project
- detailed submission requirements using **any programming language**
- GitHub and Project Submission Form guidance
- final submission checklist with evaluator notes

Projects

1. **Priority vs SRTF Comparison Project**
2. **Round Robin vs SRTF Comparison Project**
3. **Round Robin vs Priority Comparison Project**
4. **SJF vs Priority Comparison Project**
5. **Round Robin vs SJF Comparison Project**

SJF vs Priority Comparison Project

Cover Sheet - Algorithm Comparison Project

Project Title	SJF vs Priority Comparison Project	**Team Number**	
Repository URL			

Team Members

No.	Student Name	Student ID	Contribution Area	Grade	Notes
1					
2					
3					
4					
5					
6					
7					

Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	SJF	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	Priority Scheduling	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Gantt charts and execution visualization	10	-	Separate readable Gantt chart for each algorithm with correct time markers and execution order.

Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	SJF selection is based on the shortest available burst time.	
☐	Priority rule is stated clearly and applied consistently.	
☐	At least one workload shows conflict between short burst time and high priority.	
☐	Comparison tables use the same dataset for both algorithms.	
☐	Conclusion explains efficiency versus urgency trade-off.	

Project-Specific Notes for Students

- Use the same workload in both algorithms.
- Assume **preemptive** SJF and Priority Scheduling.
- Discuss efficiency versus urgency when short jobs and high-priority jobs conflict.
- Explain whether either policy causes unfair delay.

Detailed Submission Description

The following requirements apply to all comparison projects.

Implementation Technology

- Programming language: **any programming language**.
- GUI technology: must be used for the interface and visualization components.

GitHub Submission Guidelines

- Repository: the full project must be submitted through GitHub in a clean repository with meaningful commits.
- The repository must include source code, README, screenshots, sample test cases, and run instructions.
- The repository should include a clear package structure and separate classes for models, scheduling logic, GUI, metrics, and utilities.

Project Submission Form Guidelines

- A Project Submission Form must be completed and submitted with the GitHub repository link, project title, team number, and student information.
- The default branch must build and run correctly at submission time.

Recommended Repository Structure

```
project-root/
  src/
    model/
    scheduler/
    metrics/
    gui/
    util/
  assets/
  screenshots/
  test-cases/
  README.md
  .gitignore
```

Submission Checklist with Notes

Done	Submission Item	Notes
☐	GitHub repository link provided	The submitted link must be accessible and point to the final repository.
☐	README included	Must contain project description, requirements, build/run steps, and team member names.
☐	ource code included	Do not submit compiled files only. Source code must be complete.
☐	Executable/runnable instructions included	List Java version, JavaFX setup, IDE or command-line execution steps.
☐	Screenshots included	Include interface screenshots and at least one Gantt chart screenshot per algorithm.
☐	Test scenarios included	Provide at least three documented test scenarios with outputs.

☐	Project Submission Form completed	Must match the repository and team information exactly.